Umesh Arampath
Midwest Bioprocessing Center
Peoria, IL 61606, USA

July 27, 2026

Prof. John Kitchin
Associate Editor
Journal of Chemical Information and Modeling
American Chemical Society

Dear Prof. Kitchin,

We are pleased to submit a further revised version of our manuscript "When Do Simple Models Win? Machine Learning Architectures for UV Absorption Prediction" (Manuscript ID: ci-2026-009433.R1) for consideration as an Article in *Journal of Chemical Information and Modeling.*

We are grateful for the positive second-round assessment and for the remaining suggestions, which we have addressed in full.

Editor's technical issues. We corrected the Table 5 recovery-rate caption so it is consistent with the manuscript (E1); regenerated Figures 5, 6, and 9 from the v3 cross-validated predictions used in Table 2 (E2); reworded the abstract and introduction so the description of Random Forest hyperparameter tuning is consistent with the Methods (E3); and regenerated Figure 10 from the same BiGRU ensemble reported in Table 6, so the per-molecule predictions in the figure and table now match exactly (E4).

Reviewer 1's suggestions. We re-verified every reviewer comment against the original decision letter and corrected the blocks that had been paraphrased; enabled numbered (S1–S21) Supporting Information sections so all cross-references resolve correctly; defined the v2/v3 dataset versions clearly at first use with descriptive names; removed inter-revision phrasing and fixed a broken citation in two SI table captions; updated the AI-assistance disclosure to reflect that AI tools assisted with editing the manuscript; created a dedicated, cleanly organised MIT-licensed repository for the code accompanying this manuscript (`https://github.com/Umesh1608/ML_UV_models_jcim`), which the Data and Software Availability section now cites; and redrew Figure 2(b) so the solute and solvent D-MPNN encoders are shown as parallel branches.

Files included in this resubmission:

- **Clean revised manuscript**: main publication file, free of any track-changes markup.
- **Marked revised manuscript**: annotated copy with all second-round changes highlighted, uploaded as "Supporting Information for Review Only."
- **Point-by-point response to reviewers**: addresses the editor's four technical issues and Reviewer 1's seven second-round suggestions.
- **Clean Supporting Information**: final SI for publication, free of markup.

As stated in the manuscript, the authors declare no competing financial interest, and this research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit

sectors. Validated ORCID iDs for the authors are provided through the submission system.

We hope these revisions render the manuscript suitable for publication in *Journal of Chemical Information and Modeling.*

Sincerely,

Umesh Arampath
Midwest Bioprocessing Center,
Peoria, IL 61606
uarampath@mwbioprocessing.com